

Caffeine improves sustained attention in young adults: a randomized controlled trial

Maria Keller, Daniel Osei, Priya Raman

Journal of Cognitive Performance, 2021. DOI: 10.9999/jcp.2021.0042

ABSTRACT

Caffeine is the most widely consumed psychoactive substance. In this randomized placebo-controlled trial we examined whether a 200 mg dose of caffeine improves sustained attention in healthy young adults. Caffeine significantly increased sustained attention scores relative to placebo. These findings support a robust short-term attentional benefit of moderate caffeine intake in young adults.

INTRODUCTION

Sustained attention underpins performance in safety-critical occupations. Prior work has reported mixed effects of caffeine across populations and doses, motivating a controlled evaluation.

METHODS

We randomized $n = 120$ healthy young adults (ages 18-30, 61 female) to 200 mg caffeine or matched placebo in a double-blind parallel design. Sustained attention was measured with the Psychomotor Vigilance Task (PVT) 45 minutes after administration. The prespecified primary outcome was the sustained attention composite score.

RESULTS

Caffeine significantly increased sustained attention scores compared with placebo (mean difference 4.2 points, 95% CI 1.5 to 6.9, $p = 0.003$). Reaction-time lapses decreased in the caffeine group ($p = 0.01$). No serious adverse events were observed.

Table 1. Sustained attention scores by treatment group.

DISCUSSION

A moderate caffeine dose produced a reliable improvement in sustained attention among young adults. Effects in other age groups were not assessed and may differ with habitual intake.

LIMITATIONS

Single dose, single session, young-adult sample only; generalization to older adults is unknown.